

Week 10

Exploratory Data Analysis on Real Datasets

Certificate IV Information Technology (Data Science & AI)

North Metropolitan TAFE

Learning Objectives

- Apply EDA (exploratory data analysis) techniques to open datasets
- Use summary statistics and visualisations to extract insight
- Identify trends, anomalies, and relationships in data
- Present EDA findings in a clear and structured format
- Prepare for industry-grade analytical practice and assessments

Agenda

1. What is EDA; Why it matters
2. Summary statistics; quick insight from numbers
3. Visualising data; patterns at a glance
4. Working with Python; hands-on code demo
5. Spotting trends and anomalies
6. Industry case; EDA in AI dataset projects
7. Student activity; Try EDA yourself
8. How to present your results
9. Assessment preparation

What is Exploratory Data Analysis (EDA)

- EDA is the process of summarising and visualising raw data to discover patterns, test assumptions, and identify errors.
- It is the essential first step in any real-world data project.
- EDA builds confidence in how data can be used or improved.

Role of EDA in AI and Data Science

- Industry uses EDA to understand new datasets before modelling or automation.
- Good EDA prevents costly mistakes in AI pipeline development.
- In large-scale projects (like LLMs and diffusion models), EDA can reveal data issues early.

Summary Statistics; The Basics

- Common statistics; mean, median, mode, minimum, maximum, range, standard deviation
- These metrics give a quick overview of spread, central tendency, and distribution
- In Python (pandas); `dataset.describe()` gives most basic stats in one call

EDA Visualisations; Seeing the Data

- Histograms; show frequency and spread of single variables
- Boxplots; highlight medians and outliers
- Scatterplots; help spot relationships and clusters between two variables
- Use visual cues to communicate complex patterns easily

Demonstration; Basic EDA in Python

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('open_dataset.csv')
print(df.describe())
df['column1'].hist()
plt.title('Histogram of Column1')
plt.show()
```

- This code loads, describes, and plots one data column.

Interpreting Data Patterns

- Look for trends; is data increasing, decreasing, or stable over time?
- Spot anomalies; are there outliers, missing values, or strange spikes?
- Identify relationships; do variables move together, or independently?

Industry Spotlight; EDA on AI-Scale Data

- Example; Cleaning and understanding 'COCO' or 'LAION-5B' image datasets before model training
- Industry teams use EDA to find duplicate images, wrong labels, or data bias early
- EDA tools are critical for creating datasets powering AI models like LLMs or image generators

Student Activity; Practice EDA

1. Download a simple open dataset from UCI or Kaggle
2. Open it in Jupyter; run `describe()` and create one plot (histogram or scatterplot)
3. Write a short summary of one trend or anomaly you see
4. Share your notebook or summary in small groups

Structuring and Presenting EDA Findings

- Use Jupyter Notebooks for combining code, plots, and notes
- Organise findings; start with basic stats, then visualisation, then interpretations
- Clarity counts; label all plots and explain findings in plain language

Common Pitfalls and Best Practices

- Do not assume data is clean or complete without checking
- Avoid cherry-picking results; show both expected and unexpected patterns
- Always document steps; this helps reproducibility and teamwork

EDA Thread; Weeks 7 to 14

- Week 7; Clean your data before analysis
- Week 8; Version and document every EDA step
- Week 9; Ensure EDA complies with ethics and policy
- Week 10; EDA forms the foundation for all reporting and coming ML projects (Weeks 11 to 14)

Assessment Preparation

- Next assessment requires EDA on a UCI open dataset
- Practice cleaning, running summary statistics, and plotting data
- Prepare to write and present clear findings in Jupyter and a written report

Key Takeaways; Next Steps

- EDA is your launchpad for all valuable data work
- Next week; Synthesis and Reporting—turning EDA insights into compelling stories and professional outputs
- Keep practising; EDA is a skill that industry expects

Further Reading; Resources

- "An Introduction to Statistical Learning" (Ch 2; Exploratory Data Analysis)
- pandas and matplotlib documentation; see official tutorials online
- Courses on HuggingFace.co and DataCamp for hands-on open dataset work

